

1. Inverse trigonometric derivatives

Recall. Let f be continuous and strictly monotone on an open interval, and differentiable at a , with $f'(a) \neq 0$. For its inverse g , at $b = f(a)$,

$$g'(b) = \frac{1}{f'(a)}, \quad (f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

where these hypotheses hold. Lecture 8 applied this to obtain

$$\frac{d}{dx} \ln x = \frac{1}{x} \quad (x > 0), \quad \frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}} \quad (-1 < x < 1).$$

Arctangent. Write $y = \arctan x$. Then $\tan y = x$ and $-\pi/2 < y < \pi/2$. Tangent is continuous and strictly increasing on that interval, and its derivative is $\sec^2 y > 0$. The inverse theorem gives

$$\frac{d}{dx} \arctan x = \frac{1}{\sec^2 y} = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2}, \quad x \in \mathbb{R}.$$

The inverse theorem applies for every $x \in \mathbb{R}$.

Arccosine. If $y = \arccos x$, then $0 \leq y \leq \pi$ and $\cos y = x$. For $-1 < x < 1$, we have $0 < y < \pi$, so $\sin y > 0$ and

$$\frac{d}{dx} \arccos x = -\frac{1}{\sin y} = -\frac{1}{\sqrt{1-x^2}}, \quad -1 < x < 1.$$

Here cosine is continuous and strictly decreasing on $(0, \pi)$, with nonzero derivative there. The identity $\arccos x = \pi/2 - \arcsin x$ gives the same formula. Both inverse sine and inverse cosine are defined at $x = \pm 1$, but their finite two-sided derivatives are defined only on $(-1, 1)$.

Example

For $F(x) = \arctan(2x)$, the chain rule gives

$$F'(x) = \frac{2}{1 + 4x^2}, \quad x \in \mathbb{R}.$$

For $G(x) = \arccos(x^2)$, its domain is $[-1, 1]$, and

$$G'(x) = -\frac{2x}{\sqrt{1-x^4}}, \quad -1 < x < 1.$$

The domain condition is imposed on the argument x^2 before applying the outer derivative formula.

2. Inverse secant and the derivative formulas

The convention from Lecture 2 is

$$\operatorname{arcsec} x = \arccos(1/x), \quad |x| \geq 1,$$

with values in $[0, \pi] \setminus \{\pi/2\}$. For $|x| > 1$, the argument $1/x$ lies in $(-1, 1)$, so the chain rule gives

$$\frac{d}{dx} \operatorname{arcsec} x = -\frac{1}{\sqrt{1-1/x^2}} \left(-\frac{1}{x^2}\right) = \frac{1}{x^2 \sqrt{1-1/x^2}}.$$

Since $\sqrt{x^2} = |x|$,

$$\sqrt{1-1/x^2} = \frac{\sqrt{x^2-1}}{|x|}, \quad \boxed{\frac{d}{dx} \operatorname{arcsec} x = \frac{1}{|x|\sqrt{x^2-1}}, \quad |x| > 1.}$$

The absolute value is required on both components of the domain. For example,

$$(\operatorname{arcsec})'(-2) = \frac{1}{2\sqrt{3}}.$$

The remaining two inverse functions are defined by

$$\operatorname{arccot} x = \frac{\pi}{2} - \arctan x \quad (x \in \mathbb{R}), \quad \operatorname{arccsc} x = \arcsin(1/x) \quad (|x| \geq 1).$$

Their ranges are $(0, \pi)$ and $[-\pi/2, 0) \cup (0, \pi/2]$, respectively. Differentiating these identities gives

$$(\operatorname{arccot})'(x) = -\frac{1}{1+x^2}, \quad (\operatorname{arccsc})'(x) = -\frac{1}{x^2 \sqrt{1-1/x^2}} = -\frac{1}{|x|\sqrt{x^2-1}}.$$

Function	Derivative	Domain of derivative
$\arcsin x$	$1/\sqrt{1-x^2}$	$-1 < x < 1$
$\arccos x$	$-1/\sqrt{1-x^2}$	$-1 < x < 1$
$\arctan x$	$1/(1+x^2)$	$x \in \mathbb{R}$
$\operatorname{arccot} x$	$-1/(1+x^2)$	$x \in \mathbb{R}$
$\operatorname{arcsec} x$	$1/(x \sqrt{x^2-1})$	$ x > 1$
$\operatorname{arccsc} x$	$-1/(x \sqrt{x^2-1})$	$ x > 1$

3. Logarithmic derivatives

For a fixed base $a > 0$, $a \neq 1$, change of base gives

$$\log_a x = \frac{\ln x}{\ln a}, \quad \boxed{\frac{d}{dx} \log_a x = \frac{1}{x \ln a}, \quad x > 0.}$$

The denominator $\ln a$ is a nonzero constant.

If u is differentiable and $u(x) > 0$, the chain rule gives

$$\boxed{\frac{d}{dx} \ln(u(x)) = \frac{u'(x)}{u(x)}}.$$

Continuity of u makes it positive in a neighborhood of any point where its value is positive, so the composition is defined there. For base a , the corresponding formula is $u'(x)/(u(x) \ln a)$.

Absolute values. The function $\ln|x|$ is defined for $x \neq 0$. For $x > 0$, its derivative is $1/x$. For $x < 0$,

$$\frac{d}{dx} \ln(-x) = \frac{-1}{-x} = \frac{1}{x}.$$

Thus

$$\boxed{\frac{d}{dx} \ln|x| = \frac{1}{x} \quad (x \neq 0),} \quad \boxed{\frac{d}{dx} \ln|u(x)| = \frac{u'(x)}{u(x)} \quad (u(x) \neq 0).}$$

For the second formula, apply the chain rule to the differentiable outer function $t \mapsto \ln|t|$ on $\mathbb{R} \setminus \{0\}$.

Example

Let

$$F(x) = \ln \left(\frac{(x-1)^2}{x+1} \right).$$

The logarithm's argument is positive exactly when $x > -1$ and $x \neq 1$. Thus

$$D_F = (-1, 1) \cup (1, \infty).$$

On this domain, the logarithm laws give

$$F(x) = 2 \ln|x-1| - \ln(x+1).$$

Differentiating,

$$\boxed{F'(x) = \frac{2}{x-1} - \frac{1}{x+1} = \frac{x+3}{x^2-1}, \quad x \in D_F.}$$

The derivative retains the domain of the original function. In particular, the absolute value in $\ln|x-1|$ allows the identity to hold also for $-1 < x < 1$.

4. Real powers and logarithmic differentiation

A fixed real exponent. For any fixed $r \in \mathbb{R}$ and $x > 0$, the exponential and logarithm laws give

$$x^r = e^{r \ln x}.$$

The right-hand side is differentiable. Applying the chain rule,

$$\boxed{\frac{d}{dx} x^r = e^{r \ln x} \frac{r}{x} = r x^{r-1}, \quad x > 0.}$$

This proves the power rule for arbitrary real exponents on the positive half-line. Particular integer or rational powers may have larger domains, which must be considered separately. For example, $(\sqrt{x})' = 1/(2\sqrt{x})$ for $x > 0$.

A variable exponent. Suppose u, v are differentiable and $u > 0$ on an interval. Define $Y = u^v = e^{v \ln u}$. This representation proves that Y is differentiable and positive. Taking logarithms gives

$$\ln Y = v \ln u, \quad \frac{Y'}{Y} = v' \ln u + v \frac{u'}{u},$$

and hence

$$\boxed{Y' = u^v \left(v' \ln u + v \frac{u'}{u} \right)}.$$

More generally, for a differentiable, nonzero function Y , $(\ln |Y|)' = Y'/Y$. Expressing $\ln |Y|$ using logarithm laws, differentiating, and multiplying by Y is called **logarithmic differentiation**. At a zero of Y , use its original expression and other differentiation rules.

Example

For $Y(x) = x^x$, $x > 0$,

$$\ln Y = x \ln x, \quad \frac{Y'}{Y} = \ln x + 1, \quad \boxed{Y' = x^x (\ln x + 1)}.$$

Example

On $x > 1$, let

$$Y(x) = \frac{(1+x^2)^3 \sqrt{x-1}}{(x+2)^2} > 0.$$

All factors are differentiable there. Then

$$\ln Y = 3 \ln(1+x^2) + \frac{1}{2} \ln(x-1) - 2 \ln(x+2),$$

so

$$\boxed{Y' = \frac{(1+x^2)^3 \sqrt{x-1}}{(x+2)^2} \left(\frac{6x}{1+x^2} + \frac{1}{2(x-1)} - \frac{2}{x+2} \right), \quad x > 1.}$$

5. Linear approximation

Let f be differentiable at a , on an open interval containing a . Its tangent-line function is

$$L(x) = f(a) + f'(a)(x - a).$$

We call L the **linearization** of f at a . For x close to a , we use $f(x) \approx L(x)$. The derivative definition gives the precise meaning of this approximation.

Theorem. There is a function ρ , defined for sufficiently small h , such that $\rho(h) \rightarrow 0$ as $h \rightarrow 0$ and

$$f(a + h) = f(a) + f'(a)h + h\rho(h).$$

Conversely, such an identity with a constant m in place of $f'(a)$ implies that f is differentiable at a , with $f'(a) = m$.

Proof. For $h \neq 0$, define

$$\rho(h) = \frac{f(a + h) - f(a)}{h} - f'(a), \quad \rho(0) = 0.$$

The derivative definition says $\rho(h) \rightarrow 0$. Rearranging proves the identity, which also holds when $h = 0$. For the converse, divide the asserted identity by $h \neq 0$ after subtracting $f(a)$, and take the limit. $\square$

Thus the error $R(h) = f(a + h) - L(a + h)$ satisfies

$$\lim_{h \rightarrow 0} \frac{|R(h)|}{|h|} = 0.$$

Equivalently, for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$0 < |h| < \delta \implies |f(a + h) - L(a + h)| < \varepsilon|h|.$$

The error becomes small compared with the change in the argument. The theorem alone does not give a numerical value of δ or a fixed range on which a requested decimal accuracy holds.

Uniqueness. Suppose $P(a + h) = A + Bh$ also satisfies $[f(a + h) - P(a + h)]/h \rightarrow 0$. The numerator then tends to zero; continuity of f gives $A = f(a)$. With this value,

$$\frac{f(a + h) - P(a + h)}{h} = \frac{f(a + h) - f(a)}{h} - B \longrightarrow f'(a) - B.$$

Hence $B = f'(a)$. The tangent-line function is the unique function of this form whose error divided by h tends to zero.

6. Examples of linearization

Example

For $f(x) = \sqrt{x}$, choose $a = 4$. Then $f(4) = 2$, $f'(4) = 1/4$, and

$$L(x) = 2 + \frac{x-4}{4}.$$

For instance, $4.04 = 4 + 1/25$, so

$$\sqrt{4.04} \approx L(4.04) = 2 + \frac{1}{100} = \frac{201}{100} = 2.01.$$

The point $a = 4$ is chosen because the value and derivative are exact and the argument is nearby.

An error bound for this example. For $h \geq 0$, write $s = \sqrt{4+h}$ and $\ell = 2 + h/4$. Both are at least 2. Direct algebra gives

$$\ell^2 - s^2 = \left(2 + \frac{h}{4}\right)^2 - (4+h) = \frac{h^2}{16}.$$

Consequently

$$0 \leq \ell - s = \frac{h^2}{16(\ell + s)} \leq \frac{h^2}{64}.$$

At $h = 1/25$, this proves

$$0 \leq \frac{201}{100} - \sqrt{\frac{101}{25}} \leq \frac{1}{40000}.$$

Here the sign and size of the error follow from an exact algebraic identity, without a second derivative or a general error-bound theorem.

Example

For $f(x) = \ln x$ at $a = 1$,

$$f(1) = 0, \quad f'(1) = 1, \quad L(x) = x - 1.$$

Thus, for small h with $1+h > 0$,

$$\ln(1+h) \approx h, \quad \lim_{h \rightarrow 0} \frac{\ln(1+h) - h}{h} = 0.$$

For example, $\ln(101/100) \approx 1/100$. The approximation statement here comes from differentiability; no numerical bound for this particular error has been supplied.

7. Differentials

Fix a where f is differentiable. For an increment dx , define the **differential**

$$dy = f'(a) dx.$$

This is an exact definition. For $a + dx$ in the domain of f , the actual change is

$$\Delta y = f(a + dx) - f(a).$$

The remainder identity says

$$\Delta y = dy + dx \rho(dx), \quad \rho(dx) \rightarrow 0.$$

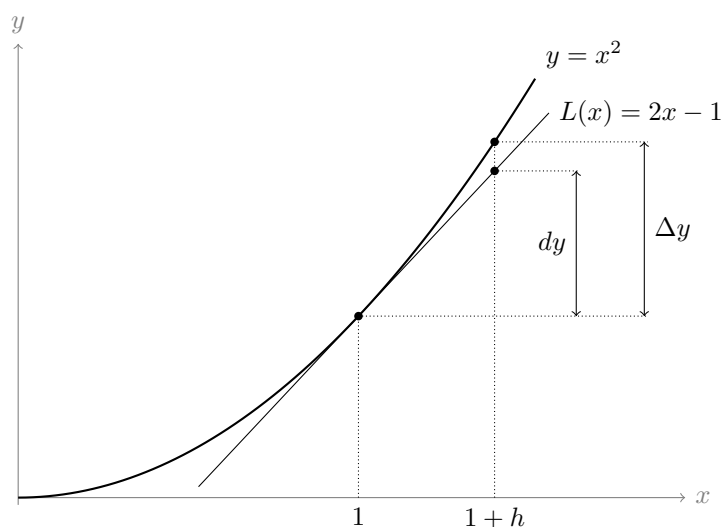
Thus dy approximates Δy for small dx . Here dx denotes an ordinary real increment; it need not be assigned any additional meaning to use this definition.

Example

For $f(x) = x^2$ at $a = 1$, let $dx = h$. Then

$$dy = 2h, \quad \Delta y = (1 + h)^2 - 1 = 2h + h^2, \quad \Delta y - dy = h^2.$$

The tangent predicts the linear term in the exact change.



At $a = 0$ for the same function, $dy = 0$ whereas $\Delta y = (dx)^2$. The theorem measures error relative to $|dx|$; it does not assert small relative error compared with $|\Delta y|$ when $f'(a) = 0$.

8. Extrema

Let f be a real-valued function on an interval I .

The function has an **absolute maximum** at $c \in I$ if

$$f(x) \leq f(c) \quad \text{for every } x \in I.$$

It has an **absolute minimum** at c if $f(x) \geq f(c)$ for every $x \in I$. The value $f(c)$ is the extremum; c is a point where it is attained. There may be several such points.

For an interior point c of I , f has a **local maximum** at c if there is $\delta > 0$ such that

$$x \in I, \quad |x - c| < \delta \quad \implies \quad f(x) \leq f(c).$$

Reverse the inequality to define a **local minimum**. Here local extrema refer to interior points; endpoints are included when discussing absolute extrema on a closed interval.

Example

For $f(x) = x^2$ on $\mathbb{R}$, zero is both a local and an absolute minimum, attained at $x = 0$. There is no absolute maximum. On $[-1, 2]$, the same formula has absolute minimum 0 at 0, and absolute maximum 4 at the endpoint 2. The domain is part of the extremum problem.

Extreme Value Theorem. If f is continuous on a closed bounded interval $[a, b]$, then there are points $c, d \in [a, b]$ such that

$$f(c) \leq f(x) \leq f(d) \quad \text{for every } x \in [a, b].$$

Thus both an absolute minimum and an absolute maximum are attained. No differentiability assumption is needed. The theorem is stated without proof.

The hypotheses. Each of the following functions is missing one of the hypotheses:

- (i) $f(x) = x$ on $(0, 1)$ is continuous, but has neither an absolute minimum nor an absolute maximum. For any x in its domain, $x/2 < x < (x+1)/2$, and both other points are in the domain.
- (ii) $f(x) = x$ on $[0, \infty)$ is continuous on a closed unbounded interval, but has no absolute maximum: $x+1$ gives a larger value than x .
- (iii) On the closed bounded interval $[0, 1]$, let

$$f(x) = \begin{cases} x, & 0 \leq x < 1, \\ 0, & x = 1. \end{cases}$$

This function is discontinuous at 1. Its values form $[0, 1)$, so it has no absolute maximum. The upper bound 1 is never attained.